

Jul 04, 2023

Introduction to Computer and Network Security

Part 1

Question 1 [50 points]

In the context of TLS and the PKI, describe:

- (a) what a digital certificate is and what are its main components [10 points];
- (b) what a root CA is [5 points];
- (c) how a system can validate a certificate [5 points];
- (d) how TLS 1.2 handshake works by drawing the message sequence chart and briefly describing each step [25 points];
- (e) how TLS ensures authentication, confidentiality, and integrity [5 points].

Part 2

Question 2 [40 points]

In the context of data protection and privacy, (a) give two possible definitions of privacy and briefly comment on them, (b) explain the linkage attack on anonymized dataset with an example, and (c) explain the trade-off between anonymity and utility in the context of realizing data sets.

Question 3 [10 points]

Describe a Cross-Site Scripting attack and the main mitigations that can be used to prevent it.

Please, remember...

Be sure to annotate the following information on each used sheet of paper (possibly, 2 sheets at most)

- Name
- Surname
- Student ID number
- Question number